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# Brachytherapy vs. external beam radiotherapy for choroidal melanoma: Survival and patterns-of-care analyses

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**ABSTRACT**

**PURPOSE:** No modern randomized trials exist comparing external beam radiotherapy (EBRT) and plaque brachytherapy (BT) for choroidal melanoma, and the optimal treatment modality is currently unknown. This study compares the patterns of care and efficacy of EBRT vs. BT based on data in the Surveillance, Epidemiology, and End Results database.

**METHODS AND MATERIALS:** The Surveillance, Epidemiology, and End Results database was queried for patients aged 20−79 diagnosed with choroidal melanoma from 2004 to 2011, treated with EBRT or BT; included patients were clinically T1−T4, N0, and M0. Overall survival and cause-specific survival curves were calculated by the Kaplan−Meier method. Univariate and multivariate analyses were performed in the survival and patterns-of-care analyses.

**RESULTS:** A total of 1004 cases (380 EBRT and 624 BT) were included in the survival analysis. There was no difference in the 5-year overall survival (83.3% EBRT vs. 82.5% BT, $p = 0.69$) and 5-year cause-specific survival (88.3% EBRT vs. 88.3% BT, $p = 0.92$). In the survival analysis, older age and advanced tumor stage were predictors of increased risk of death. In the patterns-of-care analysis, later year of diagnosis and smaller tumor stage were predictors of BT use.

**CONCLUSIONS:** Advanced tumor stage and older age seem to be independent predictors for risk of death from choroidal melanoma. The use of BT favors smaller tumors and later year of diagnosis. There is no difference in survival between those treated with EBRT or BT, and the utilization of BT is increasing. 

*Keywords:* Choroidal melanoma; Uveal melanoma; SEER database; External beam radiotherapy; Eye plaque brachytherapy

## Introduction

Choroidal melanoma is the most common primary intraocular malignancy in adults. The age-adjusted incidence of choroidal melanoma is approximately 3.3 per million people in the United States (1). Certain risk factors such as Caucasian race, light eye color, fair skin, and sensitivity to sunlight are thought to be associated with its development (2). Metastases most commonly arise in the liver, followed by the lungs, bone, skin, and central nervous system (3). In general, overall survival (OS) and cause-specific survival (CSS) are favorable but vary widely based on factors such as age, location, size, and extent of disease (4).

The staging of choroidal melanoma has significantly changed over the years. Historically, the Collaborative Ocular Melanoma Study (COMS) had three classifications for choroidal melanoma: small, medium, and large (Table 1) (5−7). The American Joint Commission on Cancer (AJCC) recently changed the staging system for choroidal melanoma in 2010 from the AJCC sixth (Table 2) to the seventh edition, incorporating notable changes into the T-staging (i.e., changes to the size criteria for T1-T4, ciliary body involvement, and amount of episcleral extension) (8).

For patients with COMS small-sized choroidal melanomas, a follow-up study prospectively found that the 5-year all-cause mortality rate was 6%, supporting active surveillance for small, slow-growing tumors (5). Historically, enucleation was the standard of care for COMS medium-sized choroidal melanoma until the publication

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